```
clc;
clear;
close all;

%% FIR FILTER PARAMETERS
b = [0.0625 0.2500 0.3750 0.2500 0.0625];

Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];

n = 0:N-1;

%% 1. DIRECT REALIZATION
yd = filter(b,1,x);

%% 2. CASCADE REALIZATION
z = roots(b);

g = b(1);
yc = g*x;

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

yc = real(yc);

%% 3. SYMMETRY AND LINEAR PHASE
symErr = max(abs(b-fliplr(b)));

D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end

%% 4. FREQUENCY, PHASE AND GROUP DELAY
[H,w] = freqz(b,1,1024);

phase = unwrap(angle(H));

gd = grpdelay(b,1,1024);

%% 5. NUMERICAL COMPARISON
err = max(abs(yd-yc));
```

```
dev = err/max(abs(yd))*100;

%% DISPLAY RESULTS
fprintf('=== FIR FILTER REALIZATION ===\n\n');
```

```
=== FIR FILTER REALIZATION ===
```

```
fprintf('Filter Coefficients:\n');
```

```
Filter Coefficients:
```

```
disp(b);
```

```
    0.0625    0.2500    0.3750    0.2500    0.0625
```

```
fprintf('Symmetry: %s\n',symmetry);
```

```
Symmetry: Symmetric
```

```
fprintf('Group Delay: %.2f samples\n',D);
```

```
Group Delay: 2.00 samples
```

```
fprintf('Maximum Error: %.3e\n',err);
```

```
Maximum Error: 8.327e-16
```

```
fprintf('Percentage Deviation: %.3e %%\n',dev);
```

```
Percentage Deviation: 2.220e-13 %
```

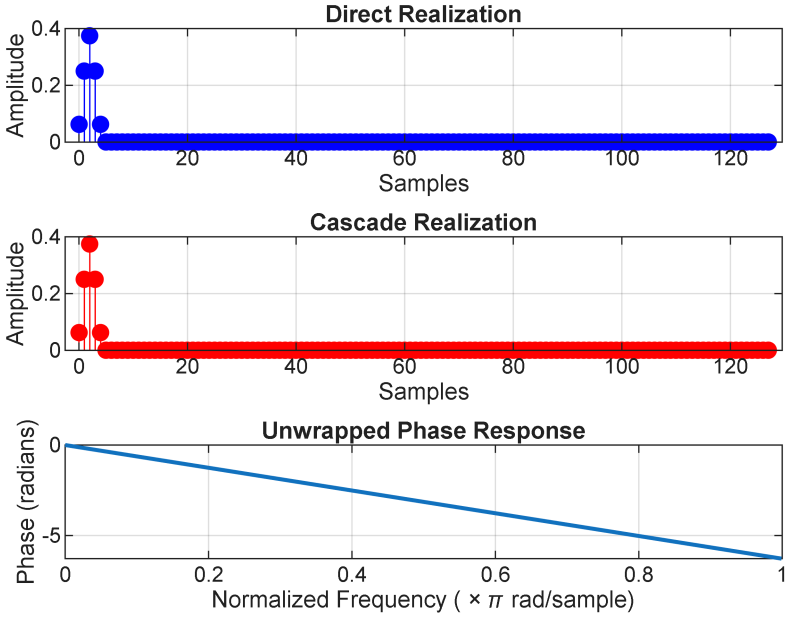
```
%% PLOTTING
figure('Name','FIR Filter Realization');

subplot(3,1,1);
stem(n,yd,'b','filled');
grid on;
title('Direct Realization');
xlabel('Samples');
ylabel('Amplitude');

subplot(3,1,2);
stem(n,yc,'r','filled');
grid on;
title('Cascade Realization');
xlabel('Samples');
ylabel('Amplitude');

subplot(3,1,3);
plot(w/pi,phase,'LineWidth',1.5);
grid on;
```

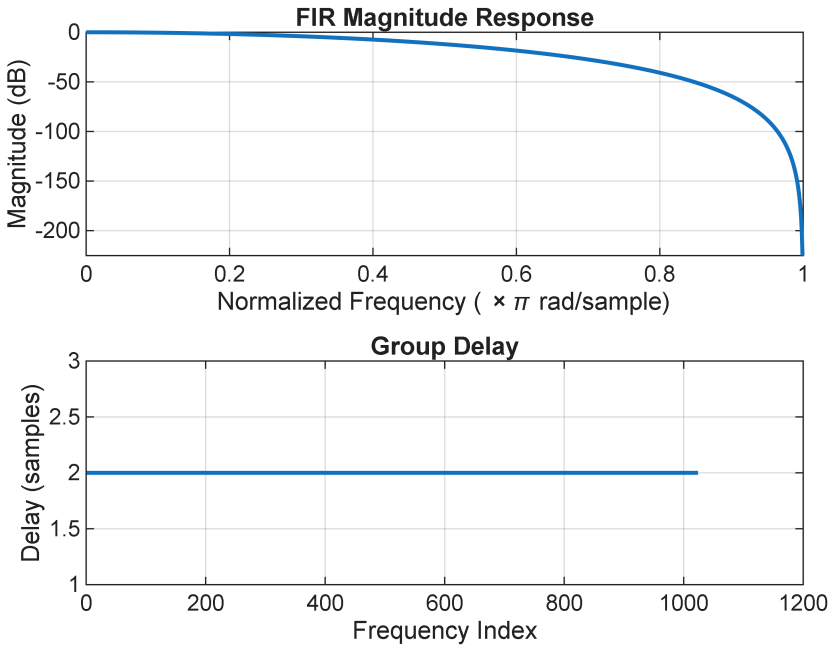
```
title('Unwrapped Phase Response');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');
```




```
figure('Name','Magnitude and Group Delay');

subplot(2,1,1);
plot(w/pi,20*log10(abs(H)),'LineWidth',1.5);
grid on;
title('FIR Magnitude Response');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Magnitude (dB)');

subplot(2,1,2);
plot(gd,'LineWidth',1.5);
grid on;
title('Group Delay');
xlabel('Frequency Index');
ylabel('Delay (samples)');
```


FIR Magnitude Response
Magnitude (dB)
0
-50
-100
-150
-200
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Group Delay
Delay (samples)
3
2.5
2
1.5
1
0
200
400
600
800
1000
1200
Frequency Index